Radical expressions

1 Write the following expressions as a single radical:

a $3\sqrt{3}$ **b** $4\sqrt{2}$ **c** $2\sqrt[3]{5}$ **d** $7\sqrt[3]{2}$

2 Complete Statement 2 by following the example in Statement 1:

a

- Statement 1: $\sqrt{9 \cdot 11} = \sqrt{9} \cdot \sqrt{11} = 3\sqrt{11}$
- Statement 2: $\sqrt{25 \cdot 2} = \sqrt{\square} \cdot \sqrt{\square} = \square\sqrt{\square}$

b

- Statement 1: $\sqrt{9 \cdot 4} = \sqrt{9} \cdot \sqrt{4} = 3 \cdot 2 = 6$
- Statement 2: $\sqrt{16 \cdot 25} = \sqrt{\square} \cdot \sqrt{\square} = \square \cdot \square = \square$

3 Simplify:

a $3\sqrt{81}$ **b** $\sqrt{49 \times 10}$ **c** $\sqrt{150}$ **d** $\dfrac{64}{\sqrt{64}}$

e $\dfrac{1}{4}\sqrt{16}$ **f** $\sqrt[3]{-1}$ **g** $\sqrt[3]{-64}$ **h** $\sqrt[3]{8}$

i $\sqrt[3]{8 \times 6}$ **j** $\sqrt[3]{81}$ **k** $7\sqrt[3]{125}$ **l** $\dfrac{1}{\sqrt[3]{8}}$

m $\dfrac{3}{2}\sqrt[3]{8}$ **n** $\sqrt[3]{108}$

4 Determine whether the following expressions are written in simplest form:

a $\sqrt{19}$ **b** $7\sqrt{50}$ **c** $\sqrt[3]{14}$

5 Simplify:

a $\sqrt{x^8}$ **b** $\sqrt[3]{m^3}$ **c** $\sqrt[3]{v^{18}}$ **d** $\sqrt[3]{6^3x^9}$

e $\sqrt[3]{b^3x^9}$ **f** $\sqrt[3]{n^9p^9}$ **g** $\sqrt[3]{9^3x^{18}y^{12}}$ **h** $\left(\sqrt[3]{7y^2}\right)^3$

i $\sqrt{a^2}$ **j** $\sqrt{b^{14}}$ **k** $\sqrt{4^2x^8}$ **l** $\sqrt{x^8y^{10}}$

m $\sqrt{j^2x^4}$ **n** $\sqrt{k^8x^8}$ **o** $\sqrt{5^2x^{14}y^{20}}$ **p** $\sqrt{28a^3y^4}$

q $\sqrt{m^4}$ **r** $\sqrt{64x}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

6 Simplify:

a $\frac{1}{3}\sqrt[3]{27 \times 4}$ **b** $\sqrt{\left(\frac{x^4}{4}\right)}$ **c** $\sqrt{\frac{49x^4}{64}}$ **d** $\sqrt{\frac{36x^{18}}{y^{20}}}$

7 Simplify:

a $\sqrt{(2x+9)^2}$ **b** $\sqrt{16y^2+24y+9}$ **c** $\sqrt{x^2y^2+18xy^2+81y^2}$

8 Evaluate the following expressions:

a $\sqrt{(-3)^2+4^2}$ **b** $\sqrt{(-10)^2-8^2}$ **c** $\sqrt{(-13)^2-(-12)^2}$

9 Evaluate the following expressions:

a $\sqrt[3]{-125} \times \sqrt[3]{27}$ **b** $\sqrt[3]{-64} \div \sqrt[3]{8}$ **c** $\sqrt[3]{-125} \times \sqrt[3]{-64}$

10 Determine whether the following statements are true or false:

a $\sqrt{(-3)^2} = -\sqrt{3^2}$ **b** $\sqrt{32} \neq \sqrt{-4} \times \sqrt{-8}$

c If x is a positive number, $\sqrt{x^{16}} = x^4$.

11 Assuming that $x > -3$, Simplify the following expression:

$$\sqrt{(x+3)^2}$$

12 Solve the following equation for k:

$$\sqrt[k]{y} \times \sqrt[k]{y} \times \sqrt[k]{y} = y^{\frac{1}{2}}$$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647